

TIME SERIES ANALYSIS OF VOLCANIC ACTIVITY ON AN ACTIVE EUROPEAN VOLCANIC ISLAND: ENHANCING A SUSTAINABLE TOURISM AND BUSINESS RESILIENCE

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ABSTRACT

Santorini is a super-volcano island that accommodates millions of tourists every year. Any volcanic manifestation may affect air traffic and local business related to touristic activity in the Mediterranean Sea, including the Aegean islands and Europe. The latest volcanic eruption of Santorini volcano was in 1950, while recent activity shows that this volcano is still active. Time series of ground thermometer logging data of a high-frequency retrieved from the volcanic site of Santorini and from the meteorological station of the local region to analyze the volcanic temperature oscillations. Separation of atmospheric effect from the volcanic temperature facilitates understanding hidden signals originating from a volcanic internal activity. Wavelet analysis has shown temperature anomalies in multiple time scale data caused by atmospheric or volcanic heat fluctuations. The decomposition of the time series into multiple time scales, such as long-term, seasonal-term, and short-term components, has been applied using the Kolmogorov-Zurbenko filter to separate the signals on the data. Signal processing with spectral analysis and wavelet application provides a thorough variable-periodogram along the time period of the data. The time series decomposition detects Santorini's main periods of volcanic ground temperature oscillation. This methodology can provide a more accurate warning system in this region and enhance a resilient society against natural hazards.

BACKGROUND

- ❑ Santorini, a super-volcano island, attracts millions of tourists annually and any volcanic activity can disrupt air traffic and local businesses.
- ❑ Recent data indicates that Santorini volcano is still active, making it crucial to analyze temperature oscillations for better understanding.
- ❑ Ground thermometer logging data from the volcanic site and a local meteorological station are used to study the temperature fluctuations.
- ❑ Wavelet analysis helps identify temperature anomalies caused by atmospheric or volcanic heat fluctuations across multiple time scales.
- ❑ The Kolmogorov-Zurbenko filter (Zurbenko, 1986) separates signals in the data, enabling the detection of long-term, seasonal-term, and short-term components.
- ❑ Spectral analysis combined with wavelet application generates a variable-periodogram, revealing the main periods of volcanic ground temperature oscillation in Santorini.
- ❑ This approach contributes to the development of a more accurate warning system and enhances resilience against natural hazards in the region.
- ❑ The poster presents the results of this analysis and highlights the differences between periodogram and wavelet application in studying volcanic temperature oscillations.

METHODS

- ✓ Data Collection: Ground thermometer logging data from the volcanic site of Santorini and a meteorological station in the local region are collected for analysis (Figure 1).
- ✓ Wavelet Analysis: Wavelet analysis is applied to the volcanic temperature data to identify temperature anomalies across multiple time scales, considering both the influence of atmospheric fluctuations and volcanic heat.
- ✓ Kolmogorov-Zurbenko Filter: Utilized to decompose the time series into long-term, seasonal-term, and short-term components, effectively separating the signals associated with atmospheric effects from those originating from volcanic internal activity (Tsakiri & Zurbenko, 2011; Tsakiri et al., 2014).
- ✓ Spectral Analysis: Spectral analysis is performed in conjunction with wavelet application to construct a variable-periodogram that covers the entire time period of the volcanic temperature data.
- ✓ Period Detection: The time series decomposition and variable-periodogram analysis enable the detection of the main periods of volcanic ground temperature oscillation in Santorini, providing insights into the underlying volcanic activity.
- ✓ Resilience Enhancement: The insights gained from this analysis contribute to the development of a more accurate warning system and enhance resilience against natural hazards in the Santorini region.

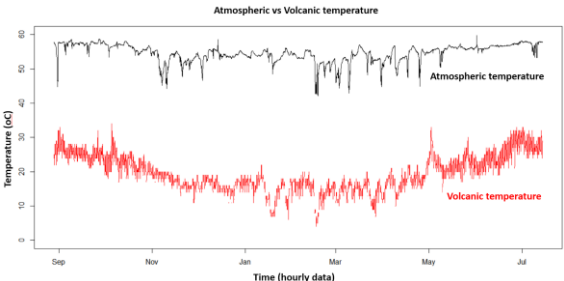


Figure 1: Raw temperature from the weather station (black line), and from the ground thermometer at Santorini volcano.

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RESULTS

Wavelet analysis

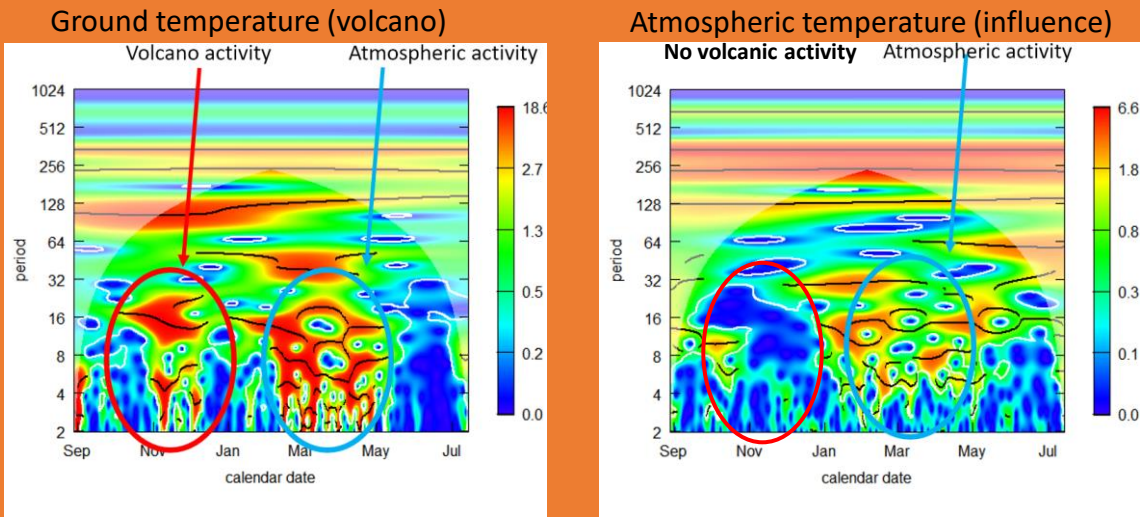


Figure 1: Wavelet analysis applied to the ground temperature data from the volcanic site of Santorini (represented by the wavelet on the left) and the atmospheric temperature data from the local weather station (represented by the wavelet on the right). The wavelets illustrate the distinct temporal characteristics and variations in temperature associated with the volcanic activity and atmospheric influences, respectively. The comparison highlights the ability of wavelet analysis to differentiate between the signals originating from the volcano and those related to atmospheric fluctuations, enabling a comprehensive understanding of the temperature dynamics in the Santorini region.

- ✓ Wavelet analysis of the ground temperature data revealed prominent volcanic activity signals, while analysis of the atmospheric temperature data showed the influence of atmospheric fluctuations.
- ✓ Decomposition and filtering techniques were applied to separate the volcanic signals from the atmospheric influences.
- ✓ The identification of volcanic temperature anomalies at various time scales contributes to the development of a reliable detection system for volcanic anomalies.
- ✓ The combination of decomposition and filtering techniques enhances the accuracy of the warning system for potential volcanic activity.
- ✓ The reliable detection system promotes a resilient and prepared society, ensuring the preservation of the local environment and supporting sustainable development in the Santorini region.

CONCLUSION

- ❑ The analysis of ground temperature data using wavelet analysis has successfully identified prominent volcanic activity signals in Santorini.
- ❑ By employing decomposition and filtering techniques, the separation of volcanic signals from atmospheric influences has been achieved, enhancing the accuracy of volcanic anomaly detection.
- ❑ The combination of these methods enables the development of a reliable warning system for potential volcanic activity, contributing to a resilient and prepared society.
- ❑ The implementation of such a system promotes sustainable development in the Santorini region by mitigating risks associated with volcanic eruptions and ensuring the preservation of the local environment.